

Culminating Assignment

Section 1: What Have You Learned?

1. Rectangles

The dimensions of a rectangle that has maximum area depend on the number of sides to be fenced.

a) When fencing all four sides of a rectangle, the optimal shape is a _____.

b) When only three sides are being fenced, the relationship between the length and the width of the rectangle is $l =$ _____.

2. Square-Based Prisms

a) For a square based prism with a given volume, the minimum surface area occurs when the prism is a _____.

b) The maximum volume for a given surface area of a square-based prism always occurs when the prism is a _____.

3. Cylinders

The maximum volume for a given surface area of a cylinder occurs when the height is _____
_____.

Section 2: Applying Your Knowledge

4. Determine the dimensions of the rectangle with the maximum area for each of the following perimeters:

a) 28 mm

b) 12 cm

c) 22 m

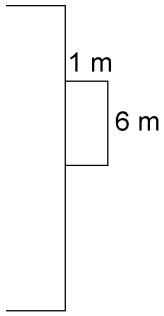
5. Find the maximum area for each rectangle in question 4.

a)

b)

c)

6. Nicole has 8 m of fencing. She plans to build a pen for her dog. The pen will have fence on three sides. The fourth side will be against the wall of her house. Nicole is considering making the pen 1 m by 6 m as shown.



- a) What are the other possible whole-number lengths for the fenced sides of the pen?

- b) Which side lengths produce the pen with the greatest area?

7. A square based prism has a volume of 512 cm^3

a) What are the dimensions of this prism that minimize the surface area?

b) What is the surface area of this prism?

8. A square based prism has a given surface area of 294 cm^2 .

a) Find the dimensions that maximize the volume

b) What is the maximum volume?

9. A cylinder has a given surface area of 750 m^2

a) What are the dimensions (radius and height) of a cylinder with maximum volume for the given surface area?

b) What is the maximum volume for the given surface area?

Extension: The Subzero Community Association is going to build a skating rink in a park. They have thirty 3 meter long sections of board that will be used to contain the rectangular rink.

a) Several different rectangular rinks are possible. Give the dimensions of two possibilities.

b) Is a square rink possible? Explain.

c) Which rink has the greatest surface area? Give its dimensions and justify your choice.

d) The ice for the rink you found in part d) is to be 3 cm thick, but it also needs to cover the wading pool that is used during the summer. This circular wading pool has a shallow cone shape with diameter 10 meters and depth 30 cm at the center. How much water will be required to initially flood this rink? Round your answer to the nearest liter.